

# ELITE IGCSE ACADEMY

*Student Past Paper Solutions*

## Jun 2023 P2H Worked Solutions

Jun 2023 | P2H

31 questions   ■   question image followed by worked solution

PREPARED BY

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**PAST PAPER QUESTION****Q#: 1****Marks: 3**TOPIC: **Algebraic Proof**

Jun 2023 P2H - Jun 2023 | P2H

1 Show that  $4\frac{2}{3} \div 1\frac{1}{5} = 3\frac{8}{9}$

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(Total for Question 1 is 3 marks)

**PAST PAPER SOLUTION****Q#: 1** **Marks: 3**TOPIC: **Algebraic Proof**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION**

Dr. Eslam Ahmed

**Topic check.** Algebraic proof. The tag is acceptable because the question asks to show the exact result.

**Solution**

$$4\frac{2}{3} = \frac{14}{3}, \quad 1\frac{1}{5} = \frac{6}{5}$$

$$4\frac{2}{3} \div 1\frac{1}{5} = \frac{14}{3} \div \frac{6}{5}$$

$$= \frac{14}{3} \times \frac{5}{6} = \frac{70}{18} = \frac{35}{9}$$

$$\frac{35}{9} = 3\frac{8}{9}$$

**FINAL ANSWER**

$$3\frac{8}{9}.$$

**PAST PAPER QUESTION****Q#: 2****Marks: 3****TOPIC: Probability Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

- 2 A biased spinner can land on green or on yellow or on brown or on pink.

The table gives the probabilities that, when the spinner is spun, it will land on green or on yellow or on brown.

| Colour      | green | yellow | brown | pink |
|-------------|-------|--------|-------|------|
| Probability | 0.32  | 0.13   | 0.28  |      |

Timucin spins the spinner 200 times.

Work out an estimate for the number of times the spinner lands on pink.

.....

**(Total for Question 2 is 3 marks)**

**PAST PAPER SOLUTION****Q#: 2****Marks: 3**TOPIC: **Probability Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Probability Toolkit. The tag is correct.**Solution**

The probability of pink is

$$1 - 0.32 - 0.13 - 0.28 = 0.27$$

Expected number of pink sweets:

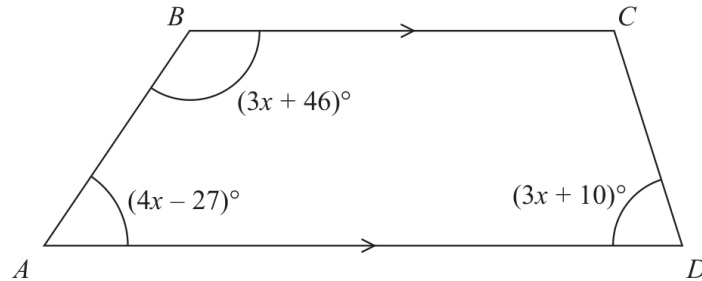
$$200 \times 0.27 = 54$$

**FINAL ANSWER**

54.

**PAST PAPER QUESTION****Q#: 3****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

**3**  $ABCD$  is a trapezium.Diagram **NOT**  
accurately drawn $BC$  is parallel to  $AD$ 

Find the size of the largest angle inside the trapezium.

o

**(Total for Question 3 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 3****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to Angles in Polygons & Parallel Lines. The question uses co-interior angles on parallel lines.

**Solution**

Since  $BC \parallel AD$ , the co-interior angles on side  $AB$  add to  $180^\circ$ :

$$(4x - 27) + (3x + 46) = 180$$

$$7x + 19 = 180$$

$$7x = 161$$

$$x = 23$$

Now find the angles:

$$\angle A = 4(23) - 27 = 65^\circ$$

$$\angle B = 3(23) + 46 = 115^\circ$$

$$\angle D = 3(23) + 10 = 79^\circ$$

Also, co-interior angles on side  $CD$  add to  $180^\circ$ :

$$\angle C = 180^\circ - 79^\circ = 101^\circ$$

The largest angle is  $115^\circ$ .

**FINAL ANSWER**

$115^\circ$ .

**PAST PAPER QUESTION****Q#: 4****Marks: 4****TOPIC: Graphs of Functions**

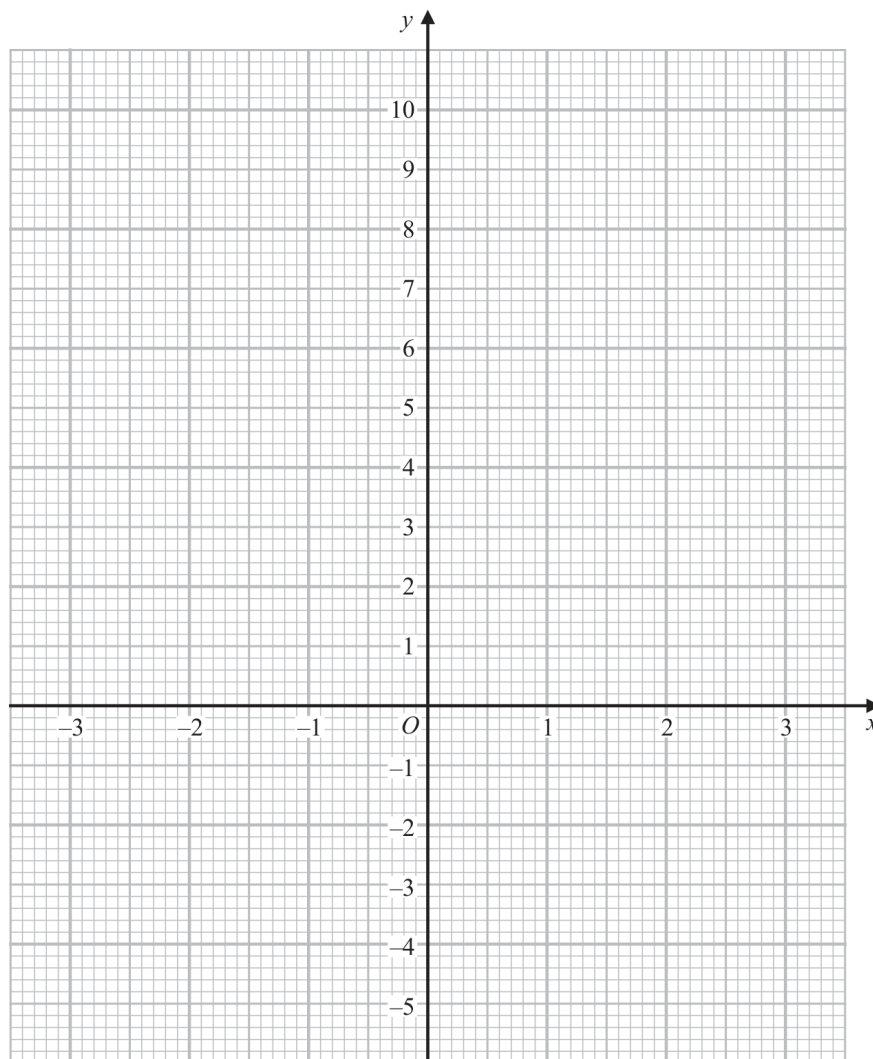
Jun 2023 P2H - Jun 2023 | P2H

- 4 (a) Complete the table of values for  $y = x^2 - x - 4$

|     |    |    |    |   |    |   |   |
|-----|----|----|----|---|----|---|---|
| $x$ | -3 | -2 | -1 | 0 | 1  | 2 | 3 |
| $y$ |    | 2  |    |   | -4 |   |   |

(2)

- (b) On the grid below, draw the graph of  $y = x^2 - x - 4$  for values of  $x$  from -3 to 3



(2)

(Total for Question 4 is 4 marks)

**PAST PAPER SOLUTION****Q#: 4****Marks: 4****TOPIC: Graphs of Functions**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct. This is a table-and-graph question for a quadratic function.**Solution**

$$y = x^2 - x - 4$$

The completed table is:

|     |    |    |    |    |    |    |   |
|-----|----|----|----|----|----|----|---|
| $x$ | -3 | -2 | -1 | 0  | 1  | 2  | 3 |
| $y$ | 8  | 2  | -2 | -4 | -4 | -2 | 2 |

Plot the points and join them with a smooth U-shaped curve.

**FINAL ANSWER**

Missing values are 8, -2, -4, -2, 2.

**PAST PAPER QUESTION****Q#: 5****Marks: 4**TOPIC: **Ratio Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

- 5 Nancy has some coins with a total value of 85 pence.  
She has only 2 pence coins and 5 pence coins.  
The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1 : 3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

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(Total for Question 5 is 4 marks)

**PAST PAPER SOLUTION****Q#: 5** **Marks: 4**TOPIC: **Ratio Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Ratio Toolkit. The tag is correct.**Method**Let the number of 2 pence coins be  $x$ .Then the number of 5 pence coins is  $3x$ .

Total value:

$$2x + 5(3x) = 85$$

$$17x = 85$$

$$x = 5$$

So there are 5 two-pence coins and 15 five-pence coins.

$$15 - 5 = 10$$

**FINAL ANSWER**

10 more 5 pence coins

**PAST PAPER QUESTION****Q#: 6****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jun 2023 P2H - Jun 2023 | P2H

- 6 (a) Write 76 000 000 in standard form.

.....  
(1)

- (b) Write  $5.4 \times 10^{-4}$  as an ordinary number.

.....  
(1)

(Total for Question 6 is 2 marks)

Dr. Eslam Ahmed

**PAST PAPER SOLUTION****Q#: 6****Marks: 2****TOPIC: Powers, Roots & Standard Form**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Powers, roots and standard form. The tag is correct.**Method**

$$76000000 = 7.6 \times 10^7$$

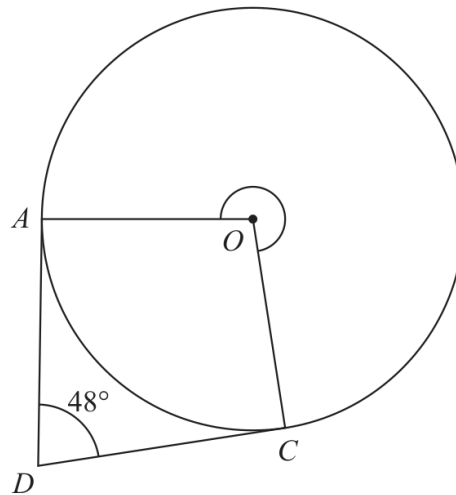
$$5.4 \times 10^{-4} = 0.00054$$

**FINAL ANSWER**(a)  $7.6 \times 10^7$ , (b) 0.00054

**PAST PAPER QUESTION****Q#: 7****Marks: 3****TOPIC: Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

7

Diagram **NOT**  
accurately drawn $A$  and  $C$  are points on a circle, centre  $O$  $DA$  is the tangent to the circle at  $A$  and  $DC$  is the tangent to the circle at  $C$ Angle  $ADC = 48^\circ$ Work out the size of reflex angle  $AOC$ 

(Total for Question 7 is 3 marks)

**PAST PAPER SOLUTION****Q#: 7****Marks: 3****TOPIC: Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**

A radius is perpendicular to a tangent, so

$$\angle OAD = 90^\circ$$

and

$$\angle OCD = 90^\circ$$

In quadrilateral  $AOCD$ ,

$$\angle AOC = 360^\circ - 90^\circ - 90^\circ - 48^\circ = 132^\circ$$

The question asks for the reflex angle:

$$360^\circ - 132^\circ = 228^\circ$$

**FINAL ANSWER**

228°.

**PAST PAPER QUESTION****Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2023 P2H - Jun 2023 | P2H

- 8 Charlotte buys a painting for \$680  
The value of the painting increases by 4% each year.
- Work out the value of the painting at the end of 3 years.  
Give your answer correct to the nearest \$

\$.....

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**(Total for Question 8 is 3 marks)**

**PAST PAPER SOLUTION****Q#: 8****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to Compound Interest & Depreciation. The value increases by the same percentage each year.

**Method**

A 4% increase each year gives a multiplier of

$$1.04$$

After 3 years, the value is

$$680(1.04)^3$$

$$= 680(1.124864)$$

$$= 764.90752$$

Correct to the nearest dollar:

$$765$$

**FINAL ANSWER****\$765**

**PAST PAPER QUESTION****Q#: 9****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2023 P2H - Jun 2023 | P2H

- 9 Change a speed of 27 kilometres per hour to a speed in metres per second.

..... m/s

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**(Total for Question 9 is 3 marks)**

Dr. Esia

**PAST PAPER SOLUTION****Q#: 9****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct. This is converting kilometres per hour to metres per second.**Solution**

To convert km/h to m/s, divide by 3.6.

$$27 \div 3.6 = 7.5$$

**FINAL ANSWER**

7.5 m/s.

**PAST PAPER QUESTION****Q#: 10****Marks: 4**TOPIC: **Statistics Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

**10** Team **A** and Team **B** take part in a quiz league.

After 11 rounds, Team **A** has a mean score per round of 17

After 9 rounds, Team **B** has a mean score per round of 18

Both teams take part in a further round.

After this round, both teams have a mean score per round of 18.5

In the further round, Team **A** scored more points than Team **B**.

How many more?

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(Total for Question 10 is 4 marks)

**PAST PAPER SOLUTION****Q#: 10****Marks: 4**TOPIC: **Statistics Toolkit**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

Team A:

$$11 \times 17 = 187$$

After the next round:

$$12 \times 18.5 = 222$$

$$\text{next score} = 222 - 187 = 35$$

Team B:

$$9 \times 18 = 162$$

After the next round:

$$10 \times 18.5 = 185$$

$$\text{next score} = 185 - 162 = 23$$

Difference:

$$35 - 23 = 12$$

**FINAL ANSWER**

12.

**PAST PAPER QUESTION****Q#: 11****Marks: 3****TOPIC: Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

- 11 Here is a 9-sided regular polygon  $ABCDEFGHIJ$ , with centre  $O$

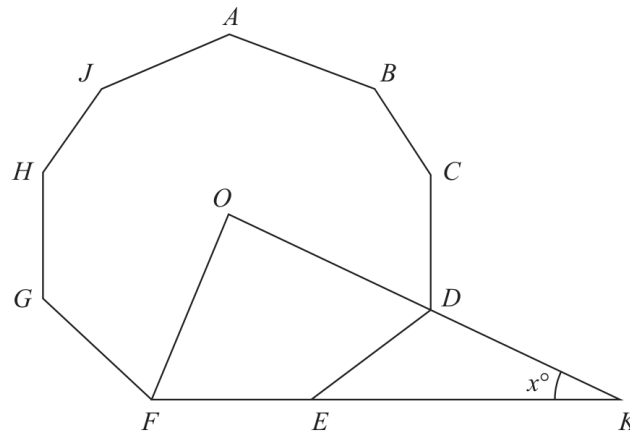


Diagram **NOT**  
accurately drawn

$ODK$  and  $FEK$  are straight lines.

Work out the value of  $x$

$x = \dots\dots\dots$

(Total for Question 11 is 3 marks)

**PAST PAPER SOLUTION****Q#: 11****Marks: 3****TOPIC: Angles in Polygons & Parallel Lines**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to angles in polygons and parallel lines. The question uses angles in a regular polygon.

**Solution**

For a regular 9-sided polygon, the angle at the centre between adjacent vertices is

$$\frac{360}{9} = 40^\circ$$

The side  $FE$  is perpendicular to the radius drawn to the midpoint of side  $FE$ .

From  $OD$  to this midpoint radius is

$$40^\circ + 20^\circ = 60^\circ$$

So the angle between  $OD$  and the side  $FE$  is

$$90^\circ - 60^\circ = 30^\circ$$

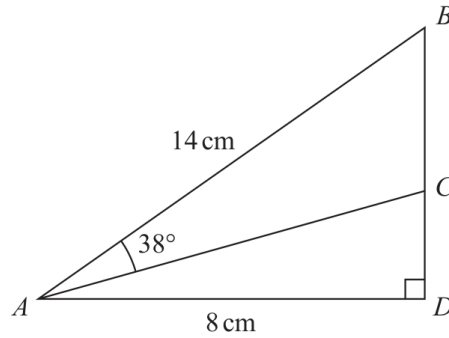
Since  $ODK$  and  $F EK$  are straight lines, this is the angle  $x$ .

**FINAL ANSWER**

$$x = 30^\circ.$$

**PAST PAPER QUESTION****Q#: 12****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2023 P2H - Jun 2023 | P2H

**12** The diagram shows right-angled triangle  $ABD$ Diagram **NOT**  
accurately drawn

$$AB = 14 \text{ cm} \quad AD = 8 \text{ cm}$$

$C$  is the point on  $BD$  such that angle  $BAC = 38^\circ$

Work out the length of  $CD$

Give your answer correct to 3 significant figures.

..... cm

**(Total for Question 12 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 12****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.**Solution**First find angle  $DAB$ :

$$\cos(\angle DAB) = \frac{8}{14}$$

$$\angle DAB = 55.150\dots^\circ$$

Given

$$\angle BAC = 38^\circ$$

So

$$\angle CAD = 55.150\dots^\circ - 38^\circ = 17.150\dots^\circ$$

In right-angled triangle  $ACD$ ,

$$\tan(17.150\dots^\circ) = \frac{CD}{8}$$

$$CD = 2.468\dots$$

**FINAL ANSWER**

2.47 cm.

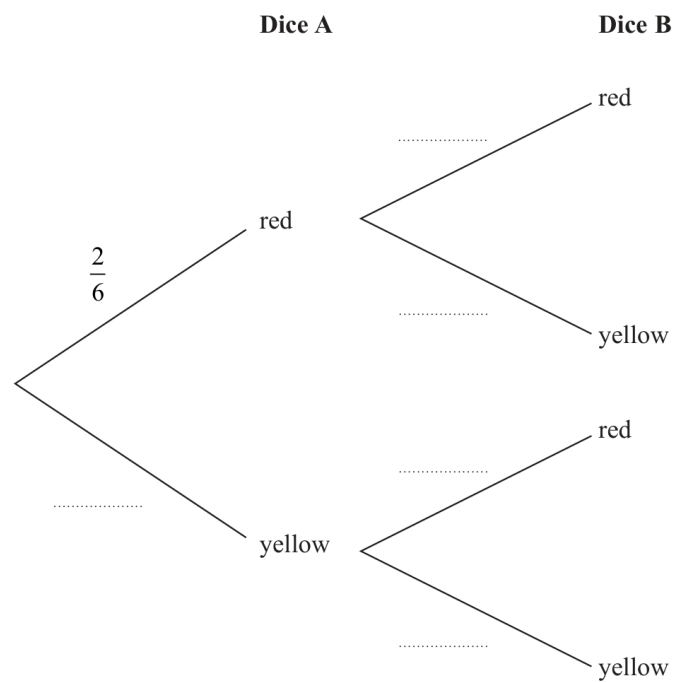
**PAST PAPER QUESTION****Q#: 13****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2023 P2H - Jun 2023 | P2H

**13** Narin has two fair 6-sided dice.Dice **A** has 2 red faces and 4 yellow faces.Dice **B** has 1 red face and 5 yellow faces.

Narin is going to throw each dice once.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both dice land on yellow.

(2)

**(Total for Question 13 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 13****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Probability Diagrams - Venn & Tree Diagrams. The tag is correct.**Solution**

Dice A has 4 yellow faces out of 6.

Dice B has 5 yellow faces out of 6.

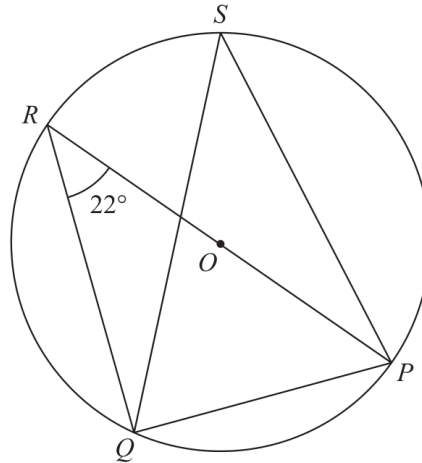
$$\begin{aligned}P(\text{both yellow}) &= \frac{4}{6} \times \frac{5}{6} \\&= \frac{5}{9}\end{aligned}$$

**FINAL ANSWER**

$$\frac{5}{9}.$$

**PAST PAPER QUESTION****Q#: 14****Marks: 4****TOPIC: Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

**14**Diagram **NOT**  
accurately drawn

$P, Q, R$  and  $S$  are points on a circle, centre  $O$   
 $ROP$  is a diameter of the circle.  
 Angle  $PRQ = 22^\circ$

(a) (i) Find the size of angle  $RQP$ .....  
(1)

(ii) Give a reason for your answer.

.....  
(1)(b) (i) Find the size of angle  $PSQ$ .....  
(1)

(ii) Give a reason for your answer.

.....  
(1)**(Total for Question 14 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 14****Marks: 4****TOPIC: Circle Theorems**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Circle Theorems. The tag is correct.**Solution**(a) Since  $RP$  is a diameter,

$$\angle RQP = 90^\circ$$

Reason: the angle in a semicircle is  $90^\circ$ .(b) Angles  $PRQ$  and  $PSQ$  stand on the same chord  $PQ$ , so they are equal.

$$\angle PSQ = 22^\circ$$

**FINAL ANSWER**

$$\angle RQP = 90^\circ, \text{ and } \angle PSQ = 22^\circ.$$

**PAST PAPER QUESTION****Q#: 15****Marks: 7****TOPIC: Algebraic Roots & Indices**

Jun 2023 P2H - Jun 2023 | P2H

15 (a) Simplify fully  $(32a^{15})^{\frac{3}{5}}$ .....  
(2)(b) Express  $\left(\frac{1}{10x}\right)^{-3}$  in the form  $px^n$  where  $p$  and  $n$  are integers......  
(2)(c) Solve  $\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$ 

Show clear algebraic working.

 $y =$  .....  
(3)**(Total for Question 15 is 7 marks)**

**PAST PAPER SOLUTION****Q#: 15****Marks: 7**TOPIC: **Algebraic Roots & Indices**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to Algebraic Roots and Indices. The first two parts are index laws; the final part is a linear equation.

**Method**

For part (a),

$$\begin{aligned}(32a^{15})^{3/5} &= 32^{3/5} a^{15 \times 3/5} \\ &= 8a^9\end{aligned}$$

For part (b),

$$\left(\frac{1}{10x}\right)^{-3} = (10x)^3 = 1000x^3$$

For part (c),

$$\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$$

Multiply by 30:

$$10(1-2y) = 24 - 15(2y-1)$$

$$10 - 20y = 39 - 30y$$

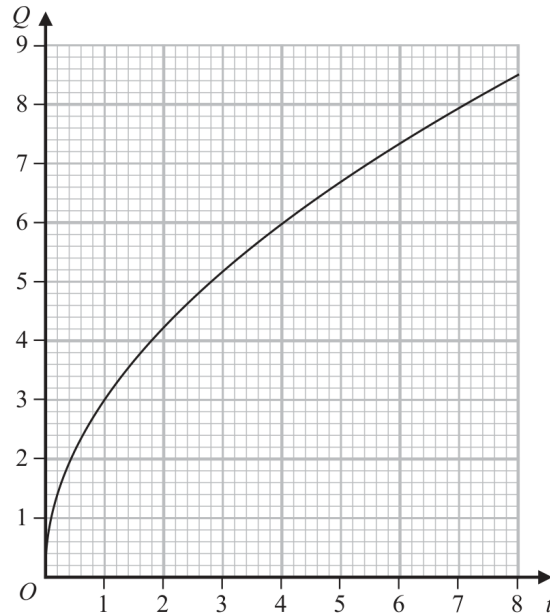
$$10y = 29$$

$$y = \frac{29}{10}$$

**FINAL ANSWER**(a)  $8a^9$ , (b)  $1000x^3$ , (c)  $y = \frac{29}{10}$

**PAST PAPER QUESTION****Q#: 16****Marks: 5****TOPIC: Direct & Inverse Proportion**

Jun 2023 P2H - Jun 2023 | P2H

**16**  $Q$  is directly proportional to  $\sqrt{t}$ The graph shows the relationship between  $Q$  and  $t$  for  $0 < t < 8$ (a) Find a formula for  $Q$  in terms of  $t$ .....  
(3) $Q$  is increased by 20%(b) Find the percentage increase in  $t$ ..... %  
(2)**(Total for Question 16 is 5 marks)**

**PAST PAPER SOLUTION****Q#: 16****Marks: 5****TOPIC: Direct & Inverse Proportion**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Corrected to Direct & Inverse Proportion. The question states  $Q$  is directly proportional to  $\sqrt{t}$ .

**Method**

$$Q \propto \sqrt{t}$$

So

$$Q = k\sqrt{t}$$

From the graph, when  $t = 4$ ,  $Q = 6$ .

$$6 = k\sqrt{4}$$

$$6 = 2k$$

$$k = 3$$

So the formula is

$$Q = 3\sqrt{t}$$

For part (b),  $Q$  is increased by 20%, so  $Q$  is multiplied by 1.2.  
Since  $Q \propto \sqrt{t}$ ,

$$t \propto Q^2$$

So  $t$  is multiplied by

$$1.2^2 = 1.44$$

This is a 44% increase.

**FINAL ANSWER**(a)  $Q = 3\sqrt{t}$ , (b) 44%

**PAST PAPER QUESTION****Q#: 17****Marks: 7****TOPIC: Rearranging Formulas**

Jun 2023 P2H - Jun 2023 | P2H

**17 (a)** Expand and simplify  $(x + 6)(3x - 2)(x + 6)$ .....  
(3)(b) Make  $e$  the subject of  $w = \sqrt{\frac{e + g}{ef - d}}$ .....  
(4)**(Total for Question 17 is 7 marks)**

**PAST PAPER SOLUTION****Q#: 17****Marks: 7****TOPIC: Rearranging Formulas**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Rearranging formulas. The tag is correct.**Solution**

(a)

$$(x + 6)(3x - 2)(x + 6) = (x + 6)^2(3x - 2)$$

$$= (x^2 + 12x + 36)(3x - 2)$$

$$= 3x^3 + 34x^2 + 84x - 72$$

(b)

$$w = \sqrt{\frac{e + g}{ef - d}}$$

$$w^2 = \frac{e + g}{ef - d}$$

$$w^2(ef - d) = e + g$$

$$e(fw^2 - 1) = g + dw^2$$

**FINAL ANSWER**

$$(a) 3x^3 + 34x^2 + 84x - 72, (b) e = \frac{g + dw^2}{fw^2 - 1}$$

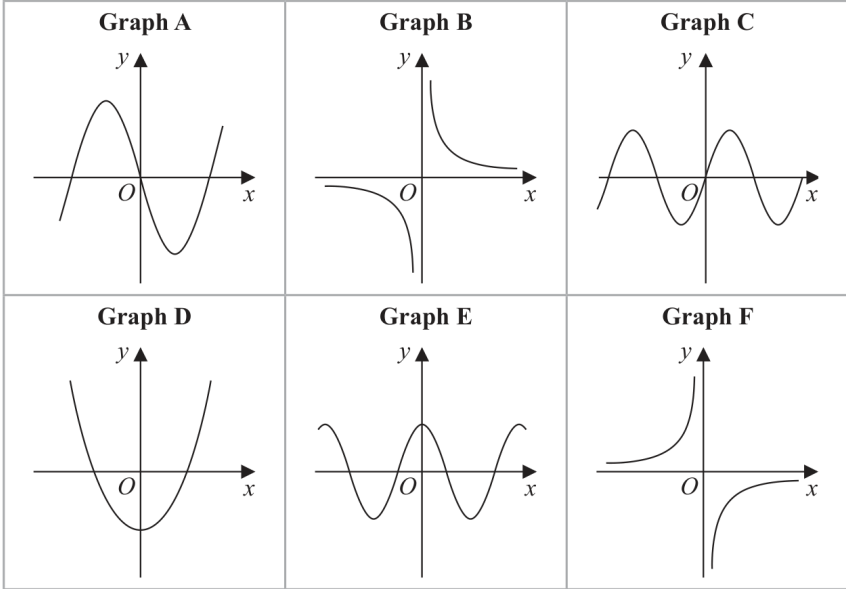
**PAST PAPER QUESTION**

**Q#: 18** **Marks: 3**

**TOPIC: Graphs of Functions**

Jun 2023 P2H - Jun 2023 | P2H

**18** Here are 6 graphs.



Complete the table below with the letter of the graph that could represent each given equation.

Write your answers on the dotted lines.

| Equation           | Graph |
|--------------------|-------|
| $y = \sin x$       | ..... |
| $y = -\frac{3}{x}$ | ..... |
| $y = 4x^3 - 5x$    | ..... |

(Total for Question 18 is 3 marks)

**PAST PAPER SOLUTION****Q#: 18****Marks: 3****TOPIC: Graphs of Functions**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Corrected to Graphs of Functions.**Solution**

$$y = \sin x$$

is the periodic sine-shaped graph, so it is **Graph C**.

$$y = \frac{3}{x}$$

is the positive reciprocal graph, so it is **Graph B**.

$$y = 4x^3 - 5x$$

is a cubic graph with two turning points, so it is **Graph A**.

**FINAL ANSWER***C, B, A.*

**PAST PAPER QUESTION****Q#: 19****Marks: 3****TOPIC: Completing the Square**

Jun 2023 P2H - Jun 2023 | P2H

**19** Express  $3x^2 - 6x + 5$  in the form  $a(x - b)^2 + c$

.....  
(Total for Question 19 is 3 marks)

**PAST PAPER SOLUTION****Q#: 19****Marks: 3****TOPIC: Completing the Square**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Completing the square. The tag is correct.**Solution**

$$3x^2 - 6x + 5 = 3(x^2 - 2x) + 5$$

$$= 3((x - 1)^2 - 1) + 5$$

$$= 3(x - 1)^2 + 2$$

**FINAL ANSWER**

$$3(x - 1)^2 + 2$$

**PAST PAPER QUESTION****Q#: 20****Marks: 3****TOPIC: Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

**20** There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

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(Total for Question 20 is 3 marks)

**PAST PAPER SOLUTION****Q#: 20****Marks: 3****TOPIC: Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

There are 3 red counters.

At least one red counter is still in the bag unless all 3 red counters are taken.

So use the complement:

$$\begin{aligned}
 P(\text{all 3 red are taken}) &= \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10} \\
 &= \frac{6}{1320} \\
 &= \frac{1}{220}
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 P(\text{at least one red is still in the bag}) &= 1 - \frac{1}{220} \\
 &= \frac{219}{220}
 \end{aligned}$$

**FINAL ANSWER**

$$\frac{219}{220}$$

**PAST PAPER QUESTION****Q#: 20****Marks: 3****TOPIC: Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

**20** There are 12 counters in a bag.

3 of the counters are red

9 of the counters are green

Ameya, Jack and Ella each take at random one counter from the bag.

Work out the probability that at least one red counter is still in the bag.

---

(Total for Question 20 is 3 marks)

**PAST PAPER SOLUTION****Q#: 20****Marks: 3****TOPIC: Combined & Conditional Probability**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

There are 3 red counters.

At least one red counter is still in the bag unless all 3 red counters are taken.

So use the complement:

$$\begin{aligned}
 P(\text{all 3 red are taken}) &= \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10} \\
 &= \frac{6}{1320} \\
 &= \frac{1}{220}
 \end{aligned}$$

Therefore,

$$\begin{aligned}
 P(\text{at least one red is still in the bag}) &= 1 - \frac{1}{220} \\
 &= \frac{219}{220}
 \end{aligned}$$

**FINAL ANSWER**

$$\frac{219}{220}$$

**PAST PAPER QUESTION****Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

**21** Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

---

(Total for Question 21 is 5 marks)

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**PAST PAPER SOLUTION****Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$2x^2 + 3y^2 = 11$$

$$x = 3y - 1$$

Substitute:

$$2(3y - 1)^2 + 3y^2 = 11$$

$$18y^2 - 12y + 2 + 3y^2 = 11$$

$$21y^2 - 12y - 9 = 0$$

$$7y^2 - 4y - 3 = 0$$

$$(7y + 3)(y - 1) = 0$$

So

$$y = -\frac{3}{7} \quad \text{or} \quad y = 1$$

Find  $x$ :

$$y = -\frac{3}{7} \Rightarrow x = -\frac{16}{7}$$

$$y = 1 \Rightarrow x = 2$$

**FINAL ANSWER**

$$(x, y) = \left(-\frac{16}{7}, -\frac{3}{7}\right) \text{ or } (2, 1)$$

**PAST PAPER QUESTION****Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

**21** Solve the simultaneous equations

$$\begin{aligned}2x^2 + 3y^2 &= 11 \\ x &= 3y - 1\end{aligned}$$

Show clear algebraic working.

---

(Total for Question 21 is 5 marks)

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**PAST PAPER SOLUTION****Q#: 21****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Correct.**Method**

$$2x^2 + 3y^2 = 11$$

$$x = 3y - 1$$

Substitute:

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$$18y^2 - 12y + 2 + 3y^2 = 11$$

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$$7y^2 - 4y - 3 = 0$$

$$(7y + 3)(y - 1) = 0$$

So

$$y = -\frac{3}{7} \quad \text{or} \quad y = 1$$

Find  $x$ :

$$y = -\frac{3}{7} \Rightarrow x = -\frac{16}{7}$$

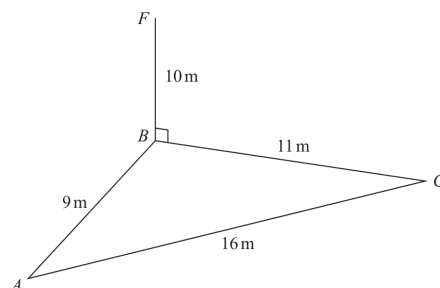
$$y = 1 \Rightarrow x = 2$$

**FINAL ANSWER**

$$(x, y) = \left(-\frac{16}{7}, -\frac{3}{7}\right) \text{ or } (2, 1)$$

**PAST PAPER QUESTION****Q#: 22** **Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2023 P2H - Jun 2023 | P2H

22 The diagram shows a triangle  $ABC$  and a flagpole  $BF$ Diagram **NOT**  
accurately drawn $A$ ,  $B$  and  $C$  are points on horizontal ground. $BF$  is vertical.

$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

 $D$  is the point on  $AC$  such that angle  $BDC = 90^\circ$ 

Work out the size of the angle of elevation of the point  $F$  from the point  $D$   
 Give your answer correct to one decimal place.

(Total for Question 22 is 5 marks)

**PAST PAPER SOLUTION****Q#: 22****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Moved to Sine, Cosine Rule & Area of Triangles.**Method**In triangle  $ABC$ ,

$$AB = 9, \quad BC = 11, \quad AC = 16$$

Use Heron's formula.

$$s = \frac{9 + 11 + 16}{2} = 18$$

$$\text{area} = \sqrt{18(18 - 9)(18 - 11)(18 - 16)}$$

$$= \sqrt{18 \cdot 9 \cdot 7 \cdot 2} = 47.6235 \dots$$

 $D$  is the foot of the perpendicular from  $B$  to  $AC$ , so  $BD$  is the height of triangle  $ABC$ .

$$\text{area} = \frac{1}{2} \times AC \times BD$$

$$47.6235 \dots = \frac{1}{2} \times 16 \times BD$$

$$BD = 5.9529 \dots$$

In right-angled triangle  $BDF$ ,

$$\tan \theta = \frac{BF}{BD}$$

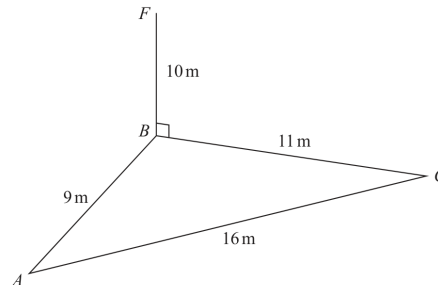
$$\tan \theta = \frac{10}{5.9529 \dots}$$

$$\theta = \tan^{-1} \left( \frac{10}{5.9529 \dots} \right) = 59.2349 \dots^\circ$$

**FINAL ANSWER****59.2°**

**PAST PAPER QUESTION****Q#: 22****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2023 P2H - Jun 2023 | P2H

22 The diagram shows a triangle  $ABC$  and a flagpole  $BF$ Diagram **NOT**  
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$$AB = 9 \text{ m} \quad BC = 11 \text{ m} \quad AC = 16 \text{ m} \quad BF = 10 \text{ m}$$

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Work out the size of the angle of elevation of the point  $F$  from the point  $D$   
 Give your answer correct to one decimal place.

(Total for Question 22 is 5 marks)

**PAST PAPER SOLUTION****Q#: 22****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Moved to Sine, Cosine Rule & Area of Triangles.**Method**In triangle  $ABC$ ,

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Use Heron's formula.

$$s = \frac{9 + 11 + 16}{2} = 18$$

$$\text{area} = \sqrt{18(18 - 9)(18 - 11)(18 - 16)}$$

$$= \sqrt{18 \cdot 9 \cdot 7 \cdot 2} = 47.6235 \dots$$

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$$\text{area} = \frac{1}{2} \times AC \times BD$$

$$47.6235 \dots = \frac{1}{2} \times 16 \times BD$$

$$BD = 5.9529 \dots$$

In right-angled triangle  $BDF$ ,

$$\tan \theta = \frac{BF}{BD}$$

$$\tan \theta = \frac{10}{5.9529 \dots}$$

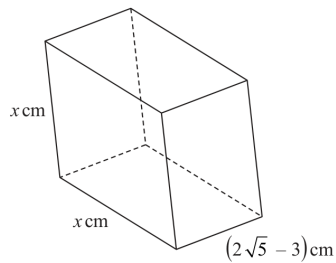
$$\theta = \tan^{-1} \left( \frac{10}{5.9529 \dots} \right) = 59.2349 \dots^\circ$$

**FINAL ANSWER** $59.2^\circ$

**PAST PAPER QUESTION****Q#: 23****Marks: 4**TOPIC: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

23 The diagram shows a cuboid with a square cross section.

Diagram **NOT**  
accurately drawnThe volume of the cuboid is  $(13 + 6\sqrt{5})\text{cm}^3$ Without using a calculator, find the value of  $x$ Give your answer in the form  $a + \sqrt{b}$  where  $a$  and  $b$  are integers.

Show your working clearly.

 $x = \dots\dots\dots$ 

(Total for Question 23 is 4 marks)

**PAST PAPER SOLUTION****Q#: 23****Marks: 4**TOPIC: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Surds. The tag is correct.**Method**

The volume is

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

So

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$$

Rationalise:

$$\begin{aligned} x^2 &= \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{(2\sqrt{5})^2 - 3^2} \\ &= \frac{26\sqrt{5} + 39 + 60 + 18\sqrt{5}}{20 - 9} \\ &= \frac{99 + 44\sqrt{5}}{11} = 9 + 4\sqrt{5} \end{aligned}$$

Since

$$(\sqrt{5} + 2)^2 = 5 + 4 + 4\sqrt{5} = 9 + 4\sqrt{5}$$

$$x = \sqrt{5} + 2$$

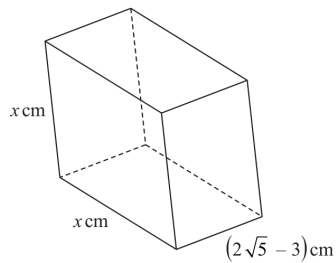
**FINAL ANSWER**

$$2 + \sqrt{5}$$

**PAST PAPER QUESTION****Q#: 23****Marks: 4**TOPIC: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

23 The diagram shows a cuboid with a square cross section.

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Show your working clearly.

 $x = \dots\dots\dots$ 

(Total for Question 23 is 4 marks)

**PAST PAPER SOLUTION****Q#: 23****Marks: 4**TOPIC: **Surds**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed***Topic check.** Corrected from Right-Angled Triangles to Surds.**Method**

The cuboid has square cross section, so its volume is

$$x^2(2\sqrt{5} - 3)$$

Given

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

So

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$$

Rationalise the denominator:

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3} \cdot \frac{2\sqrt{5} + 3}{2\sqrt{5} + 3}$$

$$x^2 = \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{20 - 9}$$

$$x^2 = \frac{99 + 44\sqrt{5}}{11}$$

$$x^2 = 9 + 4\sqrt{5}$$

Now

$$(2 + \sqrt{5})^2 = 4 + 4\sqrt{5} + 5 = 9 + 4\sqrt{5}$$

Therefore

$$x = 2 + \sqrt{5}$$

**FINAL ANSWER**

$$\boxed{2 + \sqrt{5}}$$

**PAST PAPER QUESTION****Q#: 24****Marks: 6****TOPIC: Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

**24**  $ABCD$  is a kite with  $AB = AD$  and  $CB = CD$  $A$  is the point with coordinates  $(-2, 10)$  $B$  is the point with coordinates  $\left(-\frac{27}{5}, 4\right)$  $C$  is the point with coordinates  $(4, -5)$ Work out the coordinates of  $D$ 

(....., .....)

(Total for Question 24 is 6 marks)

**PAST PAPER SOLUTION****Q#: 24****Marks: 6****TOPIC: Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION**

Dr. Eslam Ahmed

**Topic check.** Correct.**Method**

In the kite,  $AB = AD$  and  $CB = CD$ , so  $AC$  is the line of symmetry.  
Therefore  $D$  is the reflection of  $B$  in the line  $AC$ .

$$A = (-2, 10), \quad B = \left(-\frac{27}{5}, 4\right), \quad C = (4, -5)$$

A direction vector for  $AC$  is

$$\begin{pmatrix} 4 - (-2) \\ -5 - 10 \end{pmatrix} = \begin{pmatrix} 6 \\ -15 \end{pmatrix}$$

Use the simpler direction vector

$$\begin{pmatrix} 2 \\ -5 \end{pmatrix}$$

Let  $H$  be the foot of the perpendicular from  $B$  to  $AC$ .

$$\vec{AB} = \begin{pmatrix} -\frac{17}{5} \\ -6 \end{pmatrix}$$

The fraction along  $AC$  to  $H$  is

$$\begin{aligned} \frac{\vec{AB} \cdot \begin{pmatrix} 2 \\ -5 \end{pmatrix}}{2^2 + (-5)^2} &= \frac{-\frac{34}{5} + 30}{29} \\ &= \frac{\frac{116}{5}}{29} = \frac{4}{5} \end{aligned}$$

So

$$\begin{aligned} H &= A + \frac{4}{5} \begin{pmatrix} 2 \\ -5 \end{pmatrix} \\ H &= \left(-2 + \frac{8}{5}, 10 - 4\right) \\ H &= \left(-\frac{2}{5}, 6\right) \end{aligned}$$

Since  $H$  is the midpoint of  $BD$ ,

$$D = 2H - B$$

$$D = \left(-\frac{4}{5}, 12\right) - \left(-\frac{27}{5}, 4\right)$$

$$D = \left(\frac{23}{5}, 8\right)$$

**FINAL ANSWER**

$$D = \left(\frac{23}{5}, 8\right)$$

Dr. Eslam Ahmed

**PAST PAPER QUESTION****Q#: 24****Marks: 6****TOPIC: Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

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Work out the coordinates of  $D$

(....., .....)

(Total for Question 24 is 6 marks)

**PAST PAPER SOLUTION****Q#: 24****Marks: 6**TOPIC: **Coordinate Geometry**

Jun 2023 P2H - Jun 2023 | P2H

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Dr. Eslam Ahmed

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$$D = \left(\frac{23}{5}, 8\right)$$

**FINAL ANSWER**

$$D = \left(\frac{23}{5}, 8\right)$$

Dr. Eslam Ahmed

**PAST PAPER QUESTION****Q#: 25****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

- 25** A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.  
 The sphere has a mass of  $M\pi$  grams, where  $M = 260$  correct to 2 significant figures.
- Work out the upper bound for the density of the sphere.  
 Give your answer in  $\text{g/cm}^3$  correct to 2 decimal places.  
 Show your working clearly.

.....  $\text{g/cm}^3$ **(Total for Question 25 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 25****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Rounding, Estimation & Bounds is correct. The question is about choosing bounds before using the density formula.

**Method**

For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.

The radius is 2.8 cm correct to 1 decimal place, so the lower bound is

$$r = 2.75$$

$M = 260$  correct to 2 significant figures, so the upper bound is

$$M = 265$$

The mass is  $M\pi$  and the volume of the sphere is  $\frac{4}{3}\pi r^3$ .

So

$$\text{density} = \frac{M\pi}{\frac{4}{3}\pi r^3} = \frac{3M}{4r^3}$$

Upper bound:

$$\frac{3(265)}{4(2.75)^3} = 9.5567...$$

**FINAL ANSWER**

9.56 g/cm<sup>3</sup> to 2 decimal places.

**PAST PAPER QUESTION****Q#: 25****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

- 25** A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.  
 The sphere has a mass of  $M\pi$  grams, where  $M = 260$  correct to 2 significant figures.
- Work out the upper bound for the density of the sphere.  
 Give your answer in  $\text{g/cm}^3$  correct to 2 decimal places.  
 Show your working clearly.

.....  $\text{g/cm}^3$ **(Total for Question 25 is 4 marks)**

**PAST PAPER SOLUTION****Q#: 25****Marks: 4****TOPIC: Rounding, Estimation & Bounds**

Jun 2023 P2H - Jun 2023 | P2H

**WORKED SOLUTION***Dr. Eslam Ahmed*

**Topic check.** Rounding, Estimation & Bounds is correct. The question is about choosing bounds before using the density formula.

**Method**

For the upper bound of density, use the upper bound of the mass and the lower bound of the volume.

The radius is 2.8 cm correct to 1 decimal place, so the lower bound is

$$r = 2.75$$

$M = 260$  correct to 2 significant figures, so the upper bound is

$$M = 265$$

The mass is given as  $M\pi$  grams, where  $M = 265$  for the upper bound.

The volume of the sphere is

$$\frac{4}{3}\pi r^3$$

So

$$\text{density} = \frac{M\pi}{\frac{4}{3}\pi r^3}$$

The  $\pi$  cancels:

$$\text{density} = \frac{3M}{4r^3}$$

Upper bound:

$$\frac{3(265)}{4(2.75)^3} = 9.5567...$$

**FINAL ANSWER**

9.56 g/cm<sup>3</sup> to 2 decimal places.